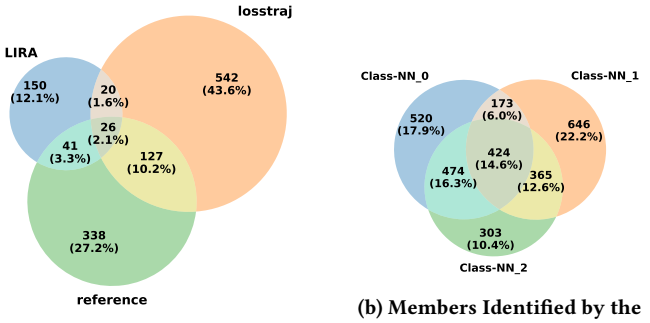


(a) Members Identified by Different Attacks

(b) Members Identified by the Same Attack's Different Instances

Figure 1: Venn diagram of member sets detected by (a) different attacks at a low FPR (0.1), (b) different instances of the Class-NN attack (with the same auxiliary dataset) at a low FPR (0.1). All Attacks are done on CIFAR-10.

Due to the critical nature of these tasks, extensive research efforts are being made to develop more effective and powerful MIAs [18]. These advances are important to ensure a more accurate and comprehensive assessment of privacy risks, auditing, and unlearning verification. While *balanced accuracy* and *AUC* are commonly used to measure the performance of MIAs, Carlini et al. [3] argue that these aggregate metrics often do not correlate with success rates at low false positive rates (FPRs), which are crucial for a practically meaningful evaluation of MIA effectiveness. Therefore, the true positive rate at low FPR (TPR@low FPR) has become the standard metric for evaluating the "practical effectiveness" of MIAs. In recent works on MIAs [3, 30, 51, 56], both aggregate metrics and TPR@low FPR are used to evaluate and demonstrate the superiority of their proposed methods over prior attacks.

In this paper, however, we argue that the evaluation, even with all these metrics, may still not capture a complete picture of MIA performance. To elaborate on this, consider a target model F_T trained on dataset $\mathcal{D}$ and two attack instances $\mathcal{A}_a$ and $\mathcal{A}_b$ having the same FPR. Suppose $\mathcal{D}_a$ and $\mathcal{D}_b$ represent the member subsets that can be detected by $\mathcal{A}_a$ and $\mathcal{A}_b$, respectively. Even if $\mathcal{A}_a$ performs better than $\mathcal{A}_b$ in both aggregate metrics and TPR@low FPR, relying only on $\mathcal{A}_a$ may not reliably assess privacy risks and verify unlearning outcomes associated with $\mathcal{D}_b \setminus \mathcal{D}_a$. For illustration, Figure 1a shows a Venn diagram of member subsets detected by three different attacks LiRA [3], Loss Trajectory [30], and Reference Attack [52], with the same FPR in our MIA experiment on CIFAR-10. The minimal overlap among them indicates that different attacks may implicitly target different subsets of members. This observation highlights a potential limitation in the common practice of favoring one attack over another in privacy-related tasks based solely on performance metrics. Better metrics do not necessarily imply a greater overall capability of an MIA, as a sample undetected by a "stronger" attack may still be exposed by another. It raises two important questions relevant to ongoing MIA privacy research and practice:

- **Q1**: Should the effectiveness of an MIA be judged solely based on those traditional metrics [10, 18]? More broadly, should research on developing new MIAs primarily focus on improving performance metrics while overlooking the member detection disparities between different methods?
- **Q2**: Is it sufficient for privacy evaluations to rely on a single "top-performing" MIA based on performance metrics without accounting for the disparities between different MIAs?

In this paper, we argue that the significant disparities in member detection at the sample level across different MIAs should not be overlooked when evaluating their effectiveness and employing them as tools for privacy assessment.

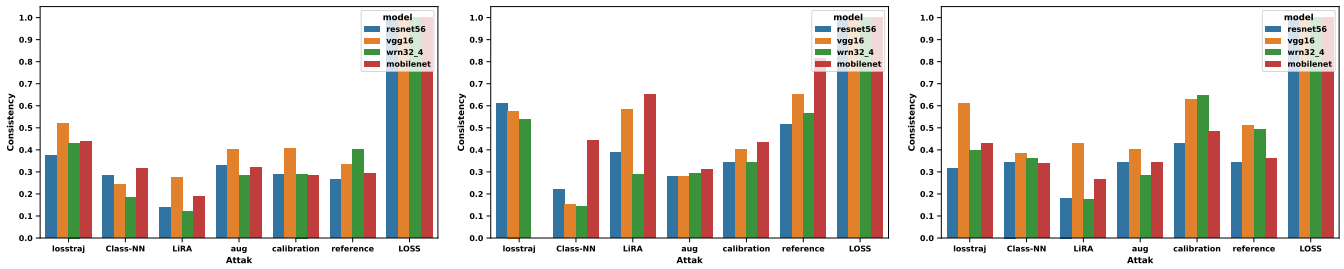
In addition, *reliability* and *consistency* are essential attributes that MIAs must possess to function effectively as privacy evaluation tools. Most existing works [10, 46, 47] that utilize MIA for privacy assessment and machine unlearning verification employ a single instance of MIA in their experiments. However, the construction of these MIAs involves inherent randomness, associated with data shuffling/sampling and training shadow/attack models, where randomness stems from factors such as optimization, weight initialization, and data batching. It has been shown that the training of randomly initialized neural networks explores different modes in the function space [12]. Therefore, factors involving randomness inevitably lead to different decision boundaries for membership detection, often resulting in significant variance in attack outcomes among different instances of the same attack with the same auxiliary knowledge. Figure 1b shows that for the same attack, Class-NN attack [44], three different instances that are trained on the same shadow dataset with different random seeds have large non-overlapping member sets. This indicates that the attack outcome can be highly sensitive to the randomness of attack construction. This raises another common issue in current research that uses MIAs for privacy assessment and performance evaluation:

- **Q3**: Is it sufficient to evaluate and report results based solely on a single instance of an MIA—as is common in existing works—while disregarding the disparities among instances that naturally arise from randomness in attack construction?

In this paper, we argue that using MIAs in their current form for evaluation, without accounting for these disparities, may lead to incomplete or potentially unreliable results.

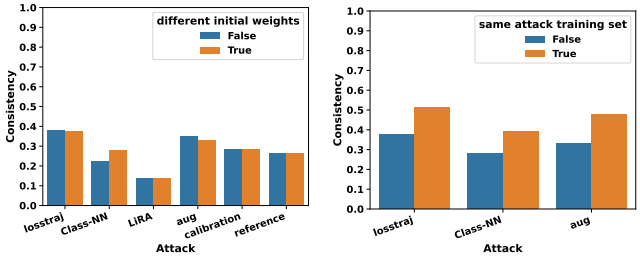
To address these concerns, this paper first systematically investigates the disparities among different MIA methods and their instances. We propose a novel framework that introduces *coverage* and *stability* analysis to evaluate and quantify the disparities of MIA methods through multiple attack instances constructed with different random seeds. Our extensive experiments highlight significant issues of instability and disparity inherent in MIAs. To better understand these disparities, we analyze the signals and features used by different MIAs to determine membership and the influence of randomness in their constructions. Our analysis reveals that different attacks may focus on samples with distinct characteristics, resulting in divergent member detection outcomes.

Furthermore, we propose an ensemble framework with three different strategies to address disparity issues in MIAs. It integrates different MIA methods from distinct perspectives: coverage-oriented, stability-oriented, and majority-oriented. These strategies combine multiple random instances of each MIA and further integrate different MIA methods to account for detection disparities. This framework not only enables the construction of more powerful attacks by leveraging the diverse strengths of existing MIAs and incorporating future advancements, but also provides an evaluation protocol to enhance the comprehensiveness of privacy evaluation.



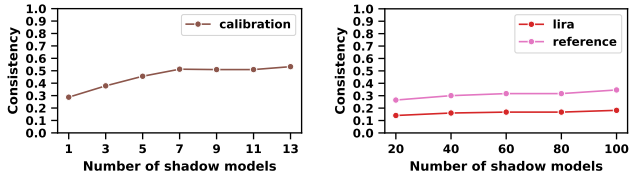
(a) CIFAR-10 (b) CIFAR-100 (c) CINIC-10

Figure 4: Consistency score shows inherent disparities among pairs of instances of MIAs (except LOSS attack) across datasets and models (ResNet-56, VGG-16, WideResNet-32, MobileNetV2). Consistency evaluated at FPR=0.1.



(a) Shadow Model Training (b) Attack Model Training

Figure 5: Shadow model training and attack model training both contribute to the disparity. Consistency is measured with six instances at FPR=0.1 on CIFAR-10.



(a) Diff-Calibration (b) LiRA and Reference Attack

Figure 6: Number of shadow models' relation with Consistency. the model architecture is ResNet-56, and the dataset is CIFAR-10. Consistency is measured with six instances at FPR=0.1.

datasets and model architectures, with an average consistency score below 0.4, highlighting the inherent instance-level inconsistency.

The instance-level MIA consistency appears to be influenced by the dataset. Attacks on CIFAR-100 demonstrate the highest overall consistency, likely due to increased over-fitting, as indicated by the larger generalization gap between the training and testing performance of target models (see the Appendix [50]). Greater over-fitting results in a larger set of common members that are easier for MIAs to infer, thereby leading to higher consistency. Additionally, certain attacks exhibit higher consistency on specific datasets and model architectures. For example, LiRA and Reference Attack achieve relatively high consistency on CIFAR-100 with VGG-16 and MobileNetV2. In contrast, Class-NN consistently shows low consistency across all datasets. This inconsistency arises from its non-overlapping shadow training sets, which lead to less aligned shadow models.

4.2.2 Disparity Factors. The randomness in MIA construction involves random partitioning of the auxiliary dataset into member and non-member sets, shadow model training, and attack model training. We investigate how these factors contribute to instance-level disparity in MIAs, particularly at low false positive rates (FPR).

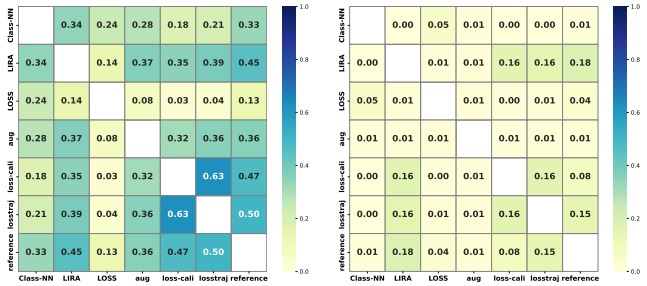
Shadow Model Training: For MIAs that rely on shadow models, shadow model training involves data shuffling and partitioning, weight initialization, and other randomness factors that are specific to an MIA, such as model distillation in Loss Trajectory Attack.

To analyze the effects of shadow model training, we compute the consistency score for instances created with different initial weights and compare it to the score for instances created with the same initial weights. When all instances start with the same set of initial weights for shadow models, data shuffling and partitioning become the primary sources of randomness for shadow model training. Figure 5a shows the consistency scores of each MIA under these two conditions. The "False" condition represents the same set of initial model weights. Comparing the two, we observe that the effect of varying initial weights on disparity is minimal (i.e., changes in the consistency score are no more than 0.05), indicating that data random shuffling and partitioning are the primary contributors to the disparity.

Attack Model Training: For attacks that include attack classification models, the training of these models can also contribute to instance-level disparity. To evaluate this effect, we fix the shadow models across all instances, ensuring that their attack models are trained on the same attack training set (corresponding to the "True" case in Figure 5b). Comparing this setup with the normal scenario where shadow models and attack training sets vary, we find that the consistency score increases by 8% to 12%. This indicates that attack model training also contributes to MIA disparity, though its impact is less pronounced than that of shadow model training.

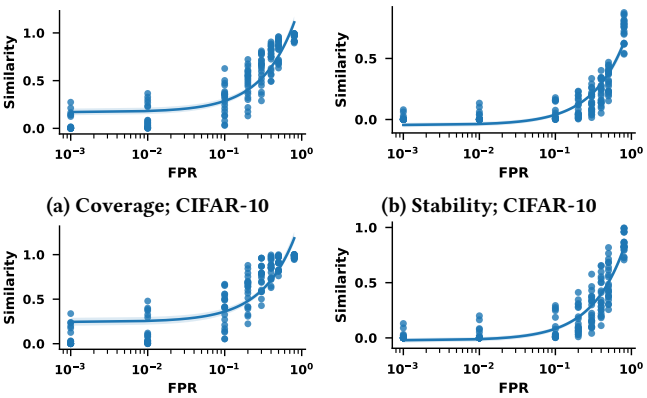
These findings highlight that both shadow model training and attack model training are critical factors driving high instance-level disparity in MIAs.

Number of Shadow Models: Additionally, we examine the impact of the number of shadow models used in an MIA on disparity. The Class-NN Attack is excluded from this analysis, as it trains shadow models on disjoint datasets, meaning that increasing the number of shadow models reduces the training data size and thus the quality of



(a) Coverage (b) Stability

Figure 10: MIA Method Disparity. The values represent the average Jaccard similarity of 4 experiment runs. Attacks' coverage and stability are calculated with six instances at FPR=0.1 on CIFAR-10.



(a) Coverage; CIFAR-10 (b) Stability; CIFAR-10

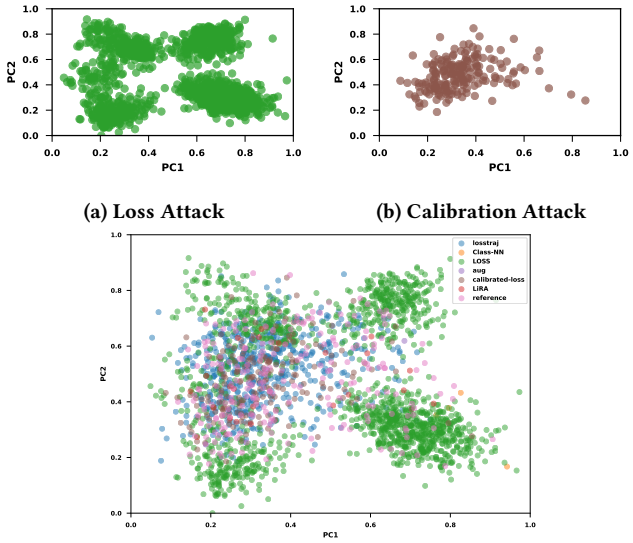
(c) Coverage; CIFAR-100 (d) Stability; CIFAR-100

Figure 11: Correlation of Disparity and FPR. The line is a linear regression on all Jaccard similarity scores for different FPR values of instances.

0.1 for stability. The notably lower Jaccard similarity in stability compared to coverage underscores the significant disparities in consistently detected vulnerabilities across different MIAs.

Certain attack pairs exhibit similar trends in both coverage and stability. For example, LiRA and Reference Attack show higher similarity in both measurements, likely due to their shared approach to shadow model training. Similarly, Loss Calibration, Loss Trajectory, and Reference Attacks show mutual similarity, likely because they are all based on loss signals. Conversely, some attack pairs show significant disagreement; for example, the Loss Attack consistently demonstrates low Jaccard similarity with all other attacks. We also observe that attacks that perform well at low FPR (e.g., Loss Trajectory, Reference attack, Loss Calibration attack, and LiRA) tend to be more mutually similar compared to attacks designed for average-case performance (e.g., Class-NN, Loss, Augmentation attack). Overall, while certain attack pairs produce membership predictions with moderately higher similarity than others, the Jaccard similarity remains low across the board when predictions are made at FPR=0.1.

We also evaluate the pairwise similarity between different MIA methods in terms of coverage and stability under varying instance-level FPR values (from 0.001 to 0.2). The results are presented in the



(a) Loss Attack (b) Calibration Attack

(c) All Attacks

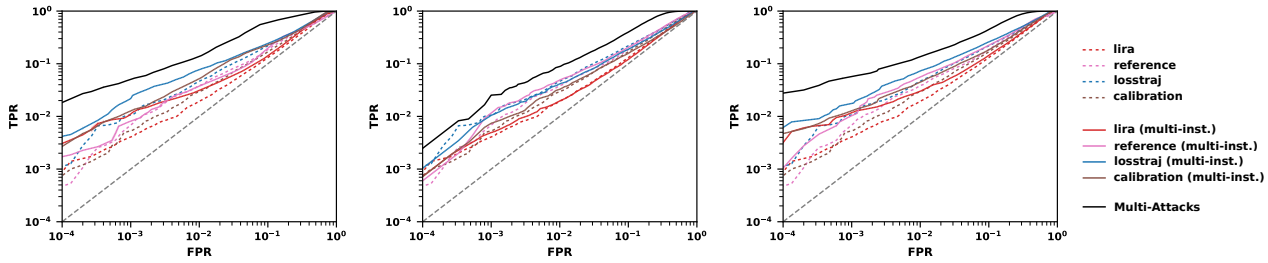
Figure 12: PCA of $\mathcal{A}$-unique Members. We conduct PCA on all the target model F_T's logits of all samples in target dataset $\mathcal{D}_T$, and plot $\mathcal{A}$-unique members (defined in Section 3.3) for each attack. It is obtained with six instances at FPR@0.1. Figure 12a and Figure 12b are picked to demonstrate the distribution difference between two MIAs. The explained variance ratio is 48.56%, and 15.95% for the PC1 (first component) and PC2 (second component), respectively.

Appendix [50]. Across different FPR settings, the overall similarity trend remains consistent, and we observe a positive correlation between FPR and similarity. In Figure 11, each point represents the Jaccard Similarity value between a pair of attacks at a given FPR level. As shown, both coverage and stability exhibit low similarity at low FPRs. As the FPR increases, the similarity also increases, with coverage similarity values approaching 1. The relatively low similarity at low FPR suggests that different attacks may have distinct insights into predicting members, especially at low FPRs. The member predictions in which these attacks are most confident tend to be nearly disjoint across methods, implying that traditional metrics—particularly those evaluated at low FPR—do not fully capture the diverse behaviors and strengths of different attacks.

4.5 Disparity Empirical Analysis

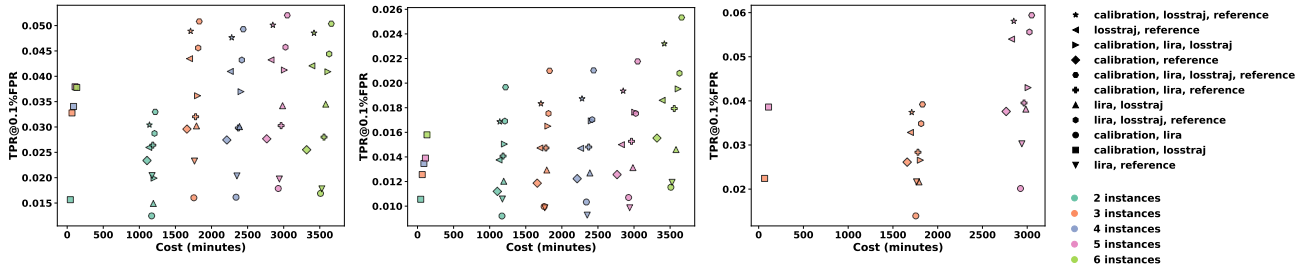
In this section, we conduct a preliminary empirical analysis to analyze the potential causes of disparities between MIAs with the methodology presented in Section 3.3. This analysis offers insights into the underlying factors that lead different MIA methods to implicitly target different subsets of training data.

4.5.1 Output Distribution of $\mathcal{A}$-Unique Samples. To understand the difference in data points that are vulnerable to different attacks, we analyze the output space of the target model F_T. We apply Principal Component Analysis (PCA) to extract feature scores from the logits predicted by F_T for each member. Focusing on $\mathcal{A}$-unique members (as defined in Section 3.3) allows us to understand the disparities among MIAs through their uniquely identified members.



(a) Stability Ensemble (b) Coverage Ensemble (c) Majority Voting Ensemble

Figure 15: ROC Curve for Ensemble. Dashed lines show single-instance ROC, solid lines show multi-instance ROC and the black line represents the complete four-attack ensemble.



(a) Stability Ensemble (b) Coverage Ensemble (c) Majority Voting Ensemble

Figure 17: Performance vs. Cost Analysis for CIFAR-10 using different ensembles.

MIA performance evaluation and development. We advocate for incorporating disparity analysis into the development and evaluation of MIAs, using our proposed coverage and stability measures to examine and quantify how an MIA differs from others in member detection. These measures offer a complementary perspective to traditional population-level metrics such as AUC and TPR@Low FPR by providing additional insight into the extent of privacy risks an attack can expose (via coverage) and the consistency with which it reveals those risks across different runs (via stability).

An MIA that achieves similar or even lower population-level metrics, such as AUC, may still hold significant value if it detects a substantially different subset of members—indicating high disparity—which can be revealed through our disparity analysis based on coverage and stability. This diversity enhances our understanding of privacy vulnerabilities by uncovering risks that other attacks may miss. Our ensemble results further support this insight, demonstrating that combining multiple attacks—including those traditionally considered "weaker"—often leads to improved overall performance. Therefore, MIAs with high disparity contribute to a more complete and robust assessment of privacy risks, especially when leveraged through our proposed ensemble strategies.

Complete and reliable privacy evaluation. As discussed in Section 4.6, the common practice of single-instance-based evaluation is insufficient and may lead to unreliable conclusions. Our ensemble framework addresses this by capturing the full spectrum of privacy risks posed by different MIA methods through multi-instance and multi-attack ensembles based on coverage, stability, and majority voting. Integrating these ensemble strategies into privacy evaluations—for instance, using ensemble attacks against defensive models— ensures a more accurate and robust assessment of privacy defenses. This approach can similarly enhance evaluations of unlearning mechanisms. Given this, it may also be necessary to revisit prior evaluations of MIA defenses and unlearning methods that relied solely on a single random MIA instance.

Additional Caveats in Using MIA for Privacy Evaluation. Recently, there has been significant interest in MIAs against large language models (LLMs). However, several works [9, 34, 57] have raised concerns about the construction of evaluation datasets, particularly regarding the distribution shift between member and non-member samples. In many LLM MIA evaluations, non-member data were collected from web content published after the model's training cutoff date, such that member samples originate from the training data distribution, while non-member samples come from a different and later distribution (e.g., different time periods). Such distribution shifts can artificially inflate MIA performance, as attacks may exploit these distribution differences rather than truly detecting membership status. Consequently, evaluations based on such datasets may overstate privacy risks.

For privacy defense evaluation, Aerni et al. [1] argues that prior evaluations using MIAs, which report attack performance averaged across all training samples, can be misleading because they may fail to reflect a defense's effectiveness against the most vulnerable examples. In addition, some evaluations have relied on relatively weak, non-adaptive attacks, potentially overstating the robustness of the proposed defenses.

Our study is orthogonal to these concerns by addressing a different overlooked issue: the instance-level and method-level disparities among MIAs, which are often neglected in current evaluation practices. Addressing these disparities requires a more holistic evaluation protocol, such as our proposed ensemble framework, to enable more complete and reliable privacy assessments.